STPM 2013 PHYSICS 960/1

Questions and Answers

All calculations use $g = 9.81\ m\ s^{-2}$ and the values of constants printed in the question paper. The retyped paper does not show part marks; the marks given here are a suggested allocation.

Section A [15 marks]

1. Ohm (Ω) is the SI unit for resistance. In terms of other units, ohm is equivalent to

A V A B J C A^{-1} C kg m^2 s^{-3} A^{-2} D kg m^2 s^{-5} A^{-2}

Answer: C

$$\Omega = \frac{V}{A} = \frac{J\,C^{-1}}{A} = \frac{kg\ m^2\ s^{-2}}{(A\ s)(A)} = kg\ m^2\ s^{-3}\ A^{-2}$$

Explanation: Use V = J/C, C = A s and J = kg m^2 s^{-2} to express the ohm in base units.

2. A long jumper jumps at an angle of 20.0° to the horizontal with a take-off speed of 11.0 m s^{-1}. What is the distance of his jump?

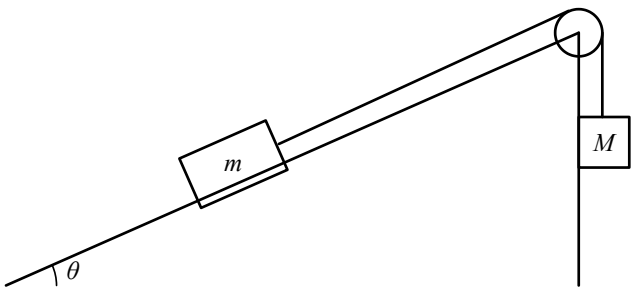
A 3.96 m B 4.22 m C 7.93 m D 12.33 m

Answer: C

$$R = \frac{u^2 \sin 2\theta}{g} = \frac{(11.0)^2 \sin 40.0°}{9.81} = 7.93\ m$$

Explanation: The jumper takes off and lands at the same level, so the range formula applies.

3. A block of mass m on a smooth plane inclined at an angle of θ to the horizontal is shown in the diagram below.



The block is connected to another block of mass M by a string over a frictionless pulley. If the block of mass m is moving up the plane, what is its acceleration?

A $\frac{g(M - m\sin\theta)}{M + m}$ B $\frac{g(m\sin\theta - M)}{M + m}$

C $\frac{g(M - m\sin\theta)}{m}$ D $\frac{g(m\sin\theta - M)}{m}$

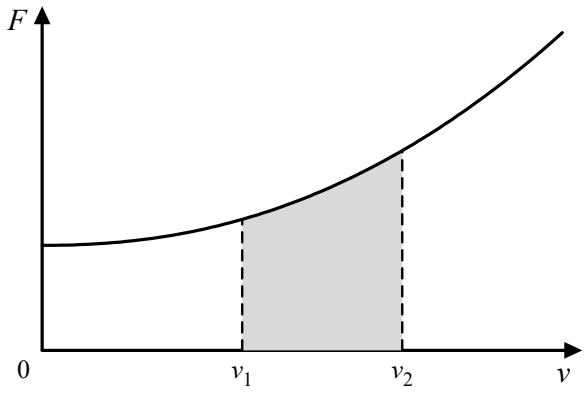
Answer: A

For M: $Mg - T = Ma$; for m: $T - mg \sin\theta = ma$

$$a = \frac{g(M - m\sin\theta)}{M + m}$$

Explanation: Treat the two blocks as one system: the net driving force is $Mg - mg \sin\theta$ and the total mass being accelerated is $M + m$.

4. The variation of the force F produced by an engine of a car with a velocity v is shown in the graph below.



When the velocity of the car increases from v_1 to v_2, the shaded area in the graph represents

A an increase in the power of the car.
B an increase in the momentum of the car.
C an increase in the work done by the car.
D an increase in the kinetic energy of the car.

Answer: A

The shaded area is $\Sigma F \Delta v$, whose unit is N × m s^{-1} = W, the unit of power. Of the options given, it corresponds to the power of the car.

Explanation: Work done would be the area under a force–**displacement** graph, and momentum change the area under a force–**time** graph. Strictly, the power at any instant is Fv; the area is chosen here because it has the dimensions of power.

5. The Mars planet moves in a circular orbit with a constant speed. The force acting on the planet is

A zero.
B parallel with the direction of its motion.
C in the direction towards the centre of the orbit.
D in the direction away from the centre of the orbit.

Answer: C

Uniform circular motion needs a resultant (centripetal) force directed towards the centre of the orbit; here it is the Sun's gravitational pull.

Explanation: The force is perpendicular to the motion, so it changes the direction but not the speed.

6. A plane flies in a vertical circle with a uniform speed of 78.0 m s^{-1}. What is the minimum radius of the circle that should be taken by the plane to ensure that its acceleration does not exceed three times the acceleration due to gravity?

A 103 m B 155 m C 207 m D 310 m

Answer: C

$$\frac{v^2}{r} \leq 3g \Rightarrow r \geq \frac{78.0^2}{3(9.81)} = 207 \text{ m}$$

Explanation: The centripetal acceleration v^2/r is larger for a smaller circle, so the limit on the acceleration gives a minimum radius.

7. A spacecraft in Earth's gravitational field moves from points L to M. The gravitational potential at L is −50 MJ kg^{-1}. Which is true of the motion and the gravitational potential energy of the spacecraft during the motion?

	Motion	*Gravitational potential energy*
A	Approaching the Earth	Increases
B	Approaching the Earth	Decreases
C	Moving away from the Earth	Increases
D	Moving away from the Earth	Decreases

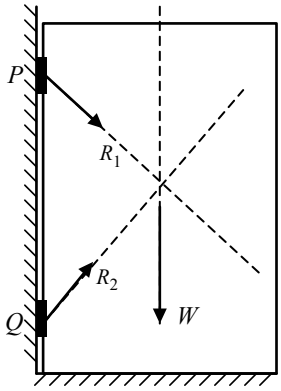
Answer: B

M is at a lower (more negative) gravitational potential than L, so the spacecraft moves closer to the Earth and its gravitational potential energy decreases.

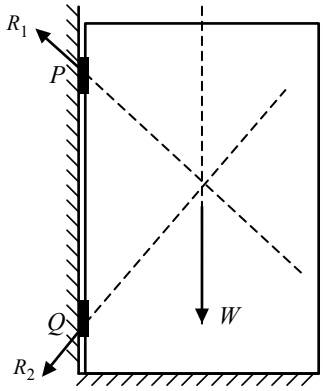
Explanation: $U = mV$: as the potential becomes more negative, the potential energy decreases. The potential is more negative closer to the Earth.

8. A door of weight W is hinged at points P and Q. The reactions of the hinges on the door at P and Q are R_1 and R_2 respectively. Which diagram shows the correct direction of the forces acting on the door?

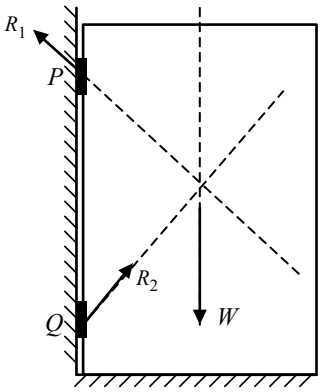
A



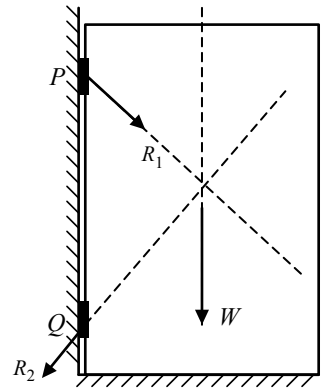
B



C



D



Answer: C

The upper hinge P pulls the door towards the wall (and up); the lower hinge Q pushes it away from the wall (and up).

For three forces in equilibrium, R_1, R_2 and W must meet at one point on the line of W, as in C.

Explanation: The weight acting at the centre tends to rotate the door away from the wall about the lower hinge, so the upper hinge must pull inwards and the lower hinge push outwards.

9. A steel cable of diameter 2.0 cm and length 10 m is used on a crane to raise a load of 1200 kg. The Young's modulus of the steel is 2.0×10^{11} Pa. Which statement about the steel cable is **not** correct?

A The cable is stretched by 1.9 mm.

B The strain energy of the cable is 3.5×10^3 J.

C The strain produced by the cable is 1.9×10^{-4}.

D The applied stress of the cable is 3.7×10^7 Pa.

Answer: B

$$F = (1200)(9.81) = 1.18 \times 10^4 \text{ N}; A = \pi(0.010)^2 = 3.14 \times 10^{-4} \text{ m}^2$$

$$\text{Stress} = 3.7 \times 10^7 \text{ Pa}; \text{strain} = 1.9 \times 10^{-4}; \text{extension} = 1.9 \text{ mm}$$

$$\text{Strain energy} = \frac{1}{2}Fe = \frac{1}{2}(1.18 \times 10^4)(1.9 \times 10^{-3}) = 11 \text{ J, not } 3.5 \times 10^3 \text{ J}$$

Explanation: Check each statement in turn. The radius is half the diameter, 1.0 cm.

10. The ideal gas equation can be derived from the kinetic theory of gases based on several assumptions. Which statement about the assumptions is **not** correct?

A The attractions between the molecules are negligible.
B The impact time is negligible compared to the time between collisions.
C The volume of the molecules is small compared to the volume of the gas.
D The collisions between the molecules and the wall of the container are inelastic.

Answer: D

Collisions with the walls are assumed to be perfectly **elastic**, not inelastic. Statement D is not correct.

Explanation: If collisions were inelastic, the molecules would lose kinetic energy and the gas would cool on its own.

11. The r.m.s. speed of the molecules of an ideal gas in a sealed container is 2.5×10^3 m s^{-1}. If the mean translational kinetic energy per molecule of the gas is 1.0×10^{-20} J, what is the mass of a molecule of the gas in the container?

A 1.6×10^{-27} kg B 3.2×10^{-27} kg C 4.0×10^{-24} kg D 8.0×10^{-24} kg

Answer: B

$$\frac{1}{2}m\langle c^2\rangle = 1.0 \times 10^{-20} \Rightarrow m = \frac{2(1.0 \times 10^{-20})}{(2.5 \times 10^3)^2} = 3.2 \times 10^{-27} \text{ kg}$$

Explanation: The mean translational kinetic energy uses the mean square speed, which is the square of the r.m.s. speed.

12. The number of degrees of freedom of a gas molecule varies according to different temperature ranges. Which number of degrees of freedom for hydrogen at the temperatures stated is correct?

	Around 30 K	*Around 300 K*	*Around 3000 K*
A	3	4	5
B	3	4	6
C	3	5	7
D	3	5	8

Answer: C

About 30 K: translation only, 3. About 300 K: + 2 rotational, 5. About 3000 K: + vibration (kinetic and potential), 7.

Explanation: Vibration contributes two degrees of freedom (kinetic and potential energy) once it becomes active at high temperatures.

13. A container of volume V contains n moles of an ideal gas at a pressure p and a temperature T. If the molar heat capacity of the gas at constant volume is $C_{V,m}$, what is the expression of the heat needed to increase the temperature of the gas by 1 K at constant pressure?

A $C_{V,m} + \frac{pV}{T}$ B $nC_{V,m} + \frac{pV}{T}$ C $C_{V,m} + \frac{npV}{T}$ D $nC_{V,m} + \frac{npV}{T}$

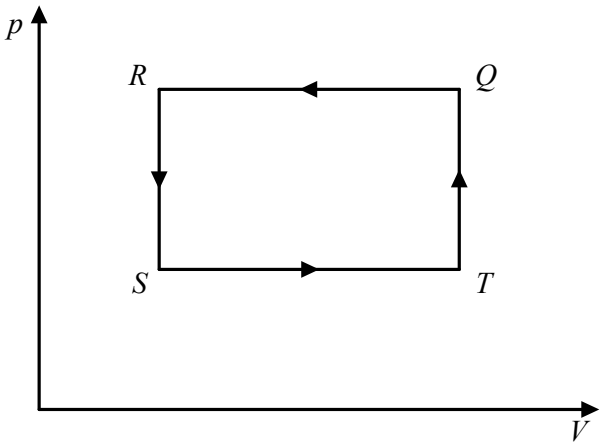
Answer: B

$$\text{Heat for 1 K at constant pressure} = nC_{p,m} = n(C_{V,m} + R) = nC_{V,m} + nR$$

$$nR = \frac{pV}{T} \Longrightarrow \text{heat} = nC_{V,m} + \frac{pV}{T}$$

Explanation: The heat is for the whole n moles, so both terms include n; the ideal gas law replaces nR by pV/T.

14. A cyclic process $QRSTQ$ for a gas is shown by the pV diagram below.



Which statement about the cyclic process $QRSTQ$ is true?

A Heat is absorbed and the internal energy of the system increases.
B Heat is released and the internal energy of the system decreases.
C Heat is absorbed and the net work is done by the system.
D Heat is released and the net work is done on the system.

Answer: D

The cycle $QRSTQ$ is anticlockwise: the gas is compressed at high pressure (QR) and expands at low pressure (ST), so net work is done **on** the system.

Over a cycle $\Delta U = 0$, so $Q = W_{by} < 0$: heat is released.

Explanation: The internal energy returns to its starting value after a complete cycle, so it neither increases nor decreases overall.

15. Which statement explains why the thermal conductivity of a metal decreases when its temperature is increased?

A The speed of its free electrons increases.
B The number of its free electrons decreases.
C The diffusion rate of its free electrons increases.
D The mean free time of its free electrons decreases.

Answer: D

At higher temperature the lattice vibrates more, so free electrons collide more often: their mean free time between collisions decreases, and they carry energy less effectively.

Explanation: This is also why the electrical resistance of a metal increases with temperature.

Section B [15 marks]

16. (a) State Newton's second law. [1]

Answer

The rate of change of momentum of a body is directly proportional to the resultant force acting on it and occurs in the direction of the force — 1

1

Explanation: In symbols, $F = \mathrm{d}p/\mathrm{d}t$.

(b) Using the above law, show that an impulse of a body is the change of momentum. [2]

Answer

$F = \frac{\mathrm{d}p}{\mathrm{d}t} \Longrightarrow F\,\mathrm{d}t = \mathrm{d}p$ — 1

Impulse $= \int F\,\mathrm{d}t = p_2 - p_1 = mv - mu$, the change of momentum — 1

2

Explanation: For a constant force this is simply $Ft = mv - mu$.

(c) A rocket has a total mass of 4.3×10^4 kg. The rate of fuel gas ejected from the rocket is 62.5 kg s^{-1} while the speed of the gas is 6.50×10^3 m s^{-1} during blast-off. Determine the

(i) thrust of the rocket, [2]

Answer

Thrust $= v\frac{\mathrm{d}m}{\mathrm{d}t} = (6.50 \times 10^3)(62.5)$ — 1

$= 4.06 \times 10^5$ N — 1

2

Explanation: The thrust is the rate of change of momentum of the ejected gas.

(ii) time taken from ignition to lift off. [2]

Answer

Lift-off when thrust = weight: $m = \frac{4.06 \times 10^5}{9.81} = 4.14 \times 10^4$ kg; fuel used = $4.30 \times 10^4 - 4.14 \times 10^4 = 1.59 \times 10^3$ kg — 1

$t = \frac{1.59 \times 10^3}{62.5} = 25.4$ s — 1

2

Total for question **7**

Explanation: At ignition the thrust is less than the weight (4.22×10^5 N), so the rocket cannot rise until enough fuel has been burnt to reduce its weight to the thrust.

17. (a) Explain the mechanisms of heat conduction. [2]

Answer

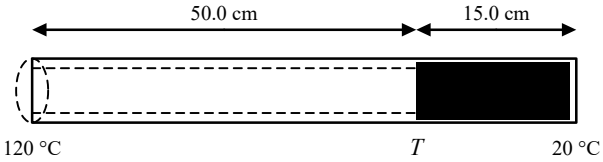
Lattice vibration: atoms at the hot end vibrate with larger amplitude and pass energy to neighbouring atoms — 1

Free electron diffusion (in metals): free electrons gain kinetic energy, diffuse to the cooler region and transfer energy by collisions with atoms — 1

2

Explanation: Metals use both mechanisms; non-metals only lattice vibration.

(b) An aluminium rod of length 50.0 cm and cross-sectional area 4.00 cm^2 is joined to a steel rod of length 15.0 cm with the same cross-sectional area as shown in the diagram below.



The side of the rod is perfectly insulated. One end of the aluminium rod is kept at 120 °C while the other end of the steel rod is kept at 20 °C. Determine

(i) the temperature at the aluminium-steel junction, [3]

Answer

Same rate in both rods: $(205)A\frac{120-T}{0.500} = (50.2)A\frac{T-20}{0.150}$ 1

$410(120-T) = 334.7(T-20)$ 1

$T = 75.1$ °C 1

3

Explanation: The sides are insulated, so in the steady state the same heat per second flows through both rods. The cross-sectional area cancels.

(ii) the total heat flow in 45.0 minutes. [3]

Answer

Rate $= (205)(4.00 \times 10^{-4})\frac{120-75.1}{0.500} = 7.37$ W 1

$Q = (7.37)(45.0 \times 60)$ 1

$Q = 1.99 \times 10^4$ J 1

3

Total for question **8**

Explanation: Convert the area to m^2 ($4.00\ cm^2 = 4.00 \times 10^{-4}\ m^2$) and the time to seconds.

[Thermal conductivity of aluminium and steel are 205 W m^{-1} K^{-1} and 50.2 W m^{-1} K^{-1} respectively.]

Section C [30 marks]

18. (a)

(i) Define work, and state the relationship between work and energy. [2]

Answer

Work is the product of the force and the displacement in the direction of the force ($W = Fs\cos\theta$) 1

Work done on an object equals the energy transferred to it (work done = change in energy) 1

2

Explanation: Work and energy have the same unit, the joule.

(ii) Using Newton's second law of motion, show that the work on an object is equal to the change in its kinetic energy. [3]

Answer

$F = ma$; $W = Fs = mas$ 1

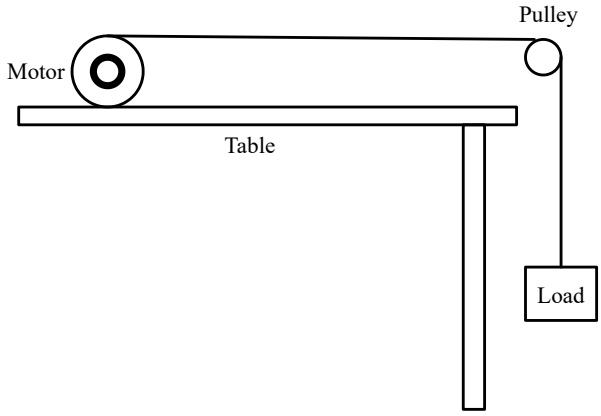
$v^2 = u^2 + 2as \Longrightarrow as = \frac{1}{2}(v^2 - u^2)$ 1

$W = \frac{1}{2}mv^2 - \frac{1}{2}mu^2 = \Delta E_k$ 1

3

Explanation: This is the work–energy theorem: the work done by the resultant force equals the change in kinetic energy.

(b) A motor pulling a load of mass 80.0 kg at rest with a constant force of 1200.0 N is shown in the diagram below.



The friction force on the pulley is 50.0 N, and the mass of the pulley is negligible. Calculate

(i) the acceleration of the load, [3]

Answer

Tension on the load = 1200.0 − 50.0 = 1150 N 1

$T - mg = ma$: $1150 - (80.0)(9.81) = 80.0a$ 1

$a = 4.57\ \text{m s}^{-2}$ 1

3

Explanation: Friction at the pulley reduces the force transmitted to the load. The load moves upward, so its weight opposes the motion.

(ii) the increase in kinetic energy of the load in 1.0 s, [2]

Answer

$v = at = (4.565)(1.0) = 4.57 \text{ m s}^{-1}$ 1

$\Delta E_k = \frac{1}{2}(80.0)(4.565)^2 = 834 \text{ J}$ 1

2

Explanation: The load starts from rest, so the increase in kinetic energy is its kinetic energy after 1.0 s.

(iii) the work done by the motor in the first second. [3]

Answer

Distance moved: $s = \frac{1}{2}at^2 = \frac{1}{2}(4.565)(1.0)^2 = 2.28 \text{ m}$ 1

$W = Fs = (1200.0)(2.28)$ 1

$W = 2.74 \times 10^3 \text{ J}$ 1

3

Explanation: The motor's force acts over the same distance as the load moves.

(c) Explain the difference between your answers in (b)(ii) and (b)(iii). [2]

Answer

The work done by the motor is greater than the gain in kinetic energy 1

because it also increases the gravitational potential energy of the load ($mgh = 1.79 \times 10^3$ J) and is used against friction at the pulley ($50.0 \times 2.28 = 114$ J) 1

2

Total for question **15**

Explanation: Check: 834 + 1791 + 114 = 2739 J, which matches the work done by the motor.

19. (a)

(i) State Newton's law of universal gravitation. [1]

Answer

Every particle attracts every other particle with a force directly proportional to the product of their masses and inversely proportional to the square of the distance between them 1

1

Explanation: $F = \frac{Gm_1m_2}{r^2}$.

(ii) Define gravitational field strength and gravitational potential. [2]

Answer

Gravitational field strength: the gravitational force per unit mass acting on a small mass placed at that point — 1

Gravitational potential: the work done per unit mass in bringing a small mass from infinity to that point — 1

2

Explanation: Field strength is a vector (N kg^{-1}); potential is a scalar (J kg^{-1}), negative everywhere.

(b) Assuming that the Earth is a uniform sphere of mass M and radius R, and that r is the distance from the centre of the Earth,

(i) determine an expression for the gravitational field strength, in terms of the universal gravitational constant G and M, at $r = R$, [2]

Answer

$mg = \frac{GMm}{R^2}$ — 1

$g = \frac{GM}{R^2}$ — 1

2

Explanation: At the surface the whole mass of the Earth acts as if concentrated at its centre.

(ii) show that the gravitational field strength is directly proportional to r for $r < R$, and it is inversely proportional to r^2 for $r > R$, [4]

Answer

Inside ($r < R$): only the mass within radius r attracts: $M' = M\frac{r^3}{R^3}$ — 1

$g = \frac{GM'}{r^2} = \frac{GM}{R^3}r \Longrightarrow g \propto r$ — 1

Outside ($r > R$): all the mass M attracts: $g = \frac{GM}{r^2}$ — 1

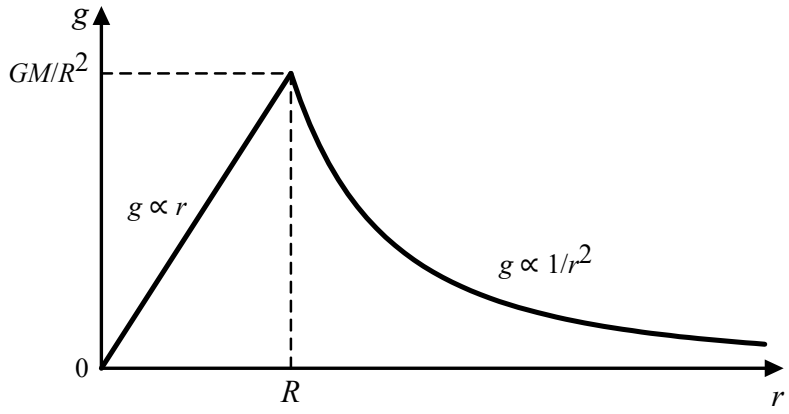
GM is constant $\Longrightarrow g \propto \frac{1}{r^2}$ — 1

4

Explanation: For a uniform sphere the mass enclosed is proportional to the volume, r^3. The shell outside radius r exerts no net force on the inside.

(iii) sketch a labelled graph of gravitational field strength against r. [2]

Answer



Straight line from the origin to a maximum GM/R^2 at $r = R$	1
Inverse-square curve decreasing towards zero for $r > R$	1
	2

Explanation: The field strength is greatest at the surface of the Earth.

(c) A space shuttle orbiting 280 km above the surface of the Earth releases a 470 kg satellite. Calculate the energy required to boost the satellite 100 km further away into another orbit. [4]

Answer

Total orbital energy $E = -\frac{GMm}{2r}$	1
$r_1 = 6.38 \times 10^6 + 2.80 \times 10^5 = 6.66 \times 10^6$ m; $r_2 = 6.76 \times 10^6$ m	1
$\Delta E = \frac{GMm}{2}\left(\frac{1}{r_1} - \frac{1}{r_2}\right) = \frac{(6.67 \times 10^{-11})(5.97 \times 10^{24})(470)}{2}\left(\frac{1}{6.66 \times 10^6} - \frac{1}{6.76 \times 10^6}\right)$	1
$\Delta E = 2.1 \times 10^8$ J	1
	4
Total for question	**15**

Explanation: The energy needed is the increase in **total** energy (kinetic + potential) between the two orbits, not just the change in potential energy. Measure both radii from the centre of the Earth.

20. (a) What do you understand by an isothermal process of an ideal gas? Derive an expression for the work done in the process. [4]

Answer

An isothermal process is one in which the temperature of the gas stays constant (so $pV =$ constant and $\Delta U = 0$) — 1

$W = \int p \, dV$ — 1

$p = \frac{nRT}{V} \Longrightarrow W = nRT\int \frac{dV}{V}$ from V_1 to V_2 — 1

$W = nRT \ln \frac{V_2}{V_1}$ — 1

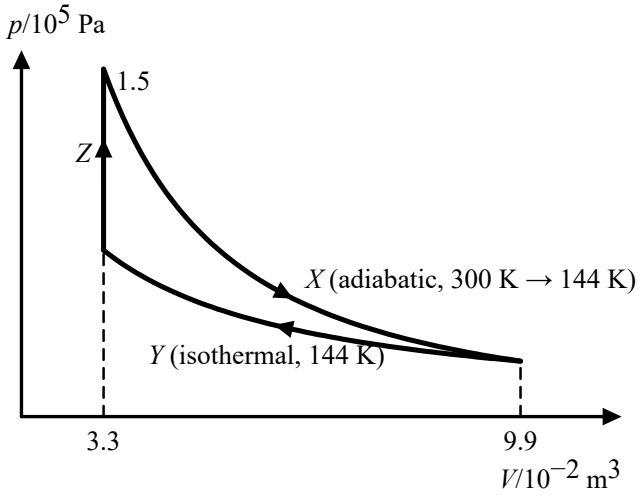
4

Explanation: Because the temperature is constant, nRT can be taken outside the integral.

(b) A cylinder with a movable, light and frictionless piston contains 2.0 moles of a monatomic ideal gas. The initial pressure, volume and temperature of the gas are 1.5×10^5 Pa, 3.3×10^{-2} m^3 and 300 K respectively. The following processes are carried out on the system.
Process X: The gas is allowed to expand adiabatically to a volume of 9.9×10^{-2} m^3.
Process Y: Then, the gas is compressed isothermally back to its initial volume.
Process Z: Finally, the gas is heated at a constant volume back to its initial state.

(i) Sketch a pV diagram to represent the above processes. [3]

Answer



X: steep adiabatic curve from (3.3, 1.5) to 9.9×10^{-2} m^3 — 1

Y: less steep isothermal curve back to 3.3×10^{-2} m^3 — 1

Z: vertical line at constant volume back to the initial state; arrows shown — 1

3

Explanation: An adiabatic curve is steeper than an isothermal curve through the same point, so the cycle closes with the vertical line Z.

(ii) Calculate the temperature of the gas after process *X*. [2]

Answer

$T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1} \Rightarrow T_2 = 300(\frac{3.3}{9.9})^{0.67}$ 1

$T_2 = 144$ K 1

2

Explanation: The gas cools as it expands adiabatically because it does work using its internal energy.

(iii) Determine the heat released in process *Y*. [3]

Answer

Isothermal: $\Delta U = 0$ so $Q = W = nRT \ln \frac{V_f}{V_i}$ 1

$Q = (2.0)(8.31)(144) \ln \frac{3.3}{9.9}$ 1

$Q = -2.6 \times 10^3$ J: 2.6×10^3 J of heat is released 1

3

Explanation: In an isothermal compression, all the work done on the gas leaves as heat.

(iv) Determine the heat absorbed in process *Z*. [3]

Answer

Constant volume: $Q = nC_{V,m}\Delta T$, $C_{V,m} = \frac{3}{2}R$ 1

$Q = (2.0)(\frac{3}{2})(8.31)(300 - 144)$ 1

$Q = 3.9 \times 10^3$ J absorbed 1

3

Total for question **15**

Explanation: No work is done at constant volume, so the heat absorbed equals the increase in internal energy as the gas returns to 300 K.

[For monatomic gas, γ is 1.67.]